MANIPAL UNIVERSITY JAIPUR

Department of Computer and Communications Engineering

<HYPNEXIS '26>

36-HOUR HACKATHON • 24–25 OCTOBER 2026

SYNORA - EVIDENCE-BACKED DATA ANALYSIS

TEAM NAME

BYTE BINDERS

TEAM ID

UV19W 67A

PROBLEM STATEMENT ID

AI-02

PROBLEM STATEMENT TITLE

DATA-ANALYST AGENT ON MESSY DATABASES

THEME / DOMAIN

AI & MACHINE LEARNING

CATEGORY

SOFTWARE

SYNORA / EVIDENCE-BACKED DATA ANALYST

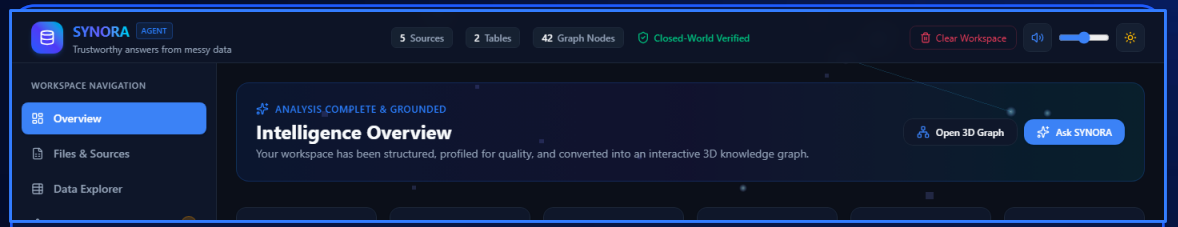
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PROBLEM STATEMENT

- Correct tables and columns are hard to discover
- Missing, duplicated, and inconsistent values distort analysis
- Natural-language questions still need safe SQL and verification
- A plausible wrong answer can be worse than no answer

OUR SOLUTION

- Schema-aware question answering
- Read-only, permission-aware query execution
- Result validation before explanation
- Evidence, provenance, and honest limitations



SYNORA workspace: analysis complete, grounded, and ready for inspection.

TECHNICAL APPROACH

SYSTEM ARCHITECTURE

Data source

↓

Schema + profile

↓

Question to SQL

↓

Policy gate

↓

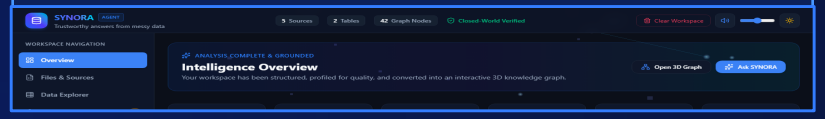
Read-only run

↓

Validate + evidence

↓

Answer / limitation



Verified workspace state

CORE TECHNOLOGIES

- React 19 + TypeScript + Vite
- Express backend with server-side OAuth routes
- Three.js / OrbitControls knowledge graph
- AlaSQL in-browser read-only analysis
- XLSX, CSV, JSON, PDF, PPTX and DOCX ingestion
- Optional Gemini endpoints for OCR / model assistance

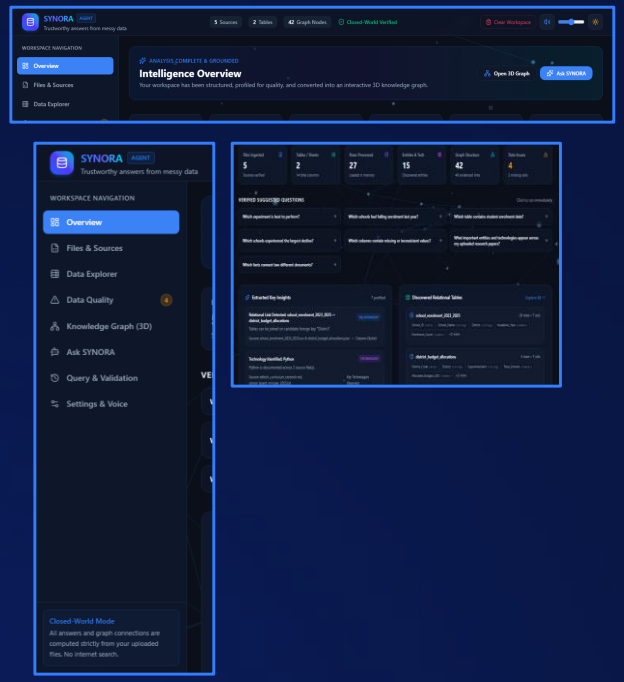
UNIQUE POINTS

- Closed-world answers grounded in uploaded data
- Query safety and role-aware access checks
- Validation records, SQL, result rows, and citations
- Explicit abstention when evidence is insufficient

USER EXPERIENCE & DESIGN

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USER INTERFACE



Workspace navigation, insights, relational tables, and persistent audio control.

USER FLOW

1. Connect or upload
2. Ask and analyse
3. Verify and understand

DESIGN GOAL

Make trustworthy data analysis understandable for people who do not write SQL.

The current prototype combines guided upload, natural-language questions, transparent query history, evidence panels, graph exploration, theme options, and persistent sound controls.



Suggested questions in the labelled demo

FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

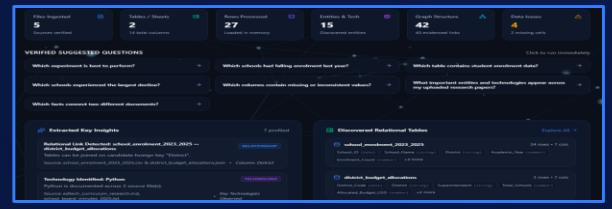
- Existing React / TypeScript application with Express entry point
- Modular ingestion, profiling, query, evidence, and graph services
- Browser-local AlaSQL keeps the demo reproducible
- Deployable as a Node server with Vite production assets
- Secrets remain server-side for optional providers

ECONOMIC FEASIBILITY

- Prototype uses open-source libraries already in the stack
- Main variable costs are optional model inference, OCR, storage, and hosting
- Early deployment can use a labelled institutional pilot workspace
- Commercial pricing is intentionally not claimed yet
- Cost validation requires real usage assumptions

OPERATIONAL VIABILITY

- Messy files and ambiguous questions remain real risks
- Mitigations: bounded queries, role checks, validation, citations, and abstention
- OAuth requires provider-console credentials and exact callbacks
- Large-document and scanned-PDF processing may need optional model services
- Monitor failures rather than masking them

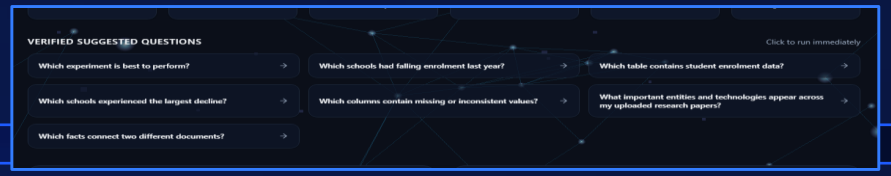


Runtime evidence: extracted insights and tables

IMPACT AND BENEFITS

1 SOCIAL IMPACT

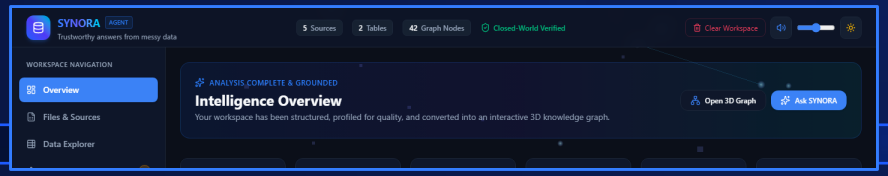
- Makes structured analysis more approachable for nontechnical users`r- Reduces dependence on manually writing SQL for supported questions`r- Encourages evidence inspection before decisions



Evidence starts with a visible question and a bounded workflow

2 ENVIRONMENTAL IMPACT

- Reuses existing organizational data`r- Favors bounded, on-demand processing`r- Avoids claiming measured savings without a lifecycle study



A grounded workspace state is visible to the user

3 TECHNICAL IMPACT

- Unifies schema discovery, execution, validation, and explanation`r- Makes query reasoning and source evidence inspectable`r- Treats missing evidence as a first-class result

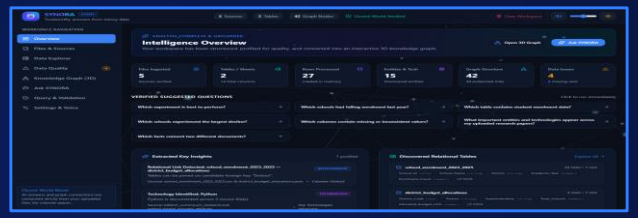
4 EDUCATIONAL IMPACT

- Shows users the SQL and result context`r- Supports guided exploration of structured data`r- Helps learners connect questions, evidence, and validation

BUSINESS MODEL & SUSTAINABILITY

TARGET USERS

- Educational institutions and academic departments
- Analysts working with spreadsheets and structured data
- Small and medium organisations with fragmented datasets
- Teams needing a safer interface to existing databases



Working SYNORA dashboard â€¢ synthetic demo

REVENUE & COST MODEL

- Free or limited pilot workspace
- Paid team workspaces
- Institutional / enterprise deployment
- Optional managed inference or document processing
- Revenue model requires validation with actual users

SUSTAINABILITY

- Reusable connectors and modular integrations
- Configurable model providers
- Fits existing organisational workflows
- Expand support from observed user requirements
- Keep the local-first demo reproducible

RESEARCH AND REFERENCES

KEY TECH & RESEARCH SOURCES

Problem context

- HYPNEXIS26 AI-02: Data-Analyst Agent on Messy Databases (supplied brief)

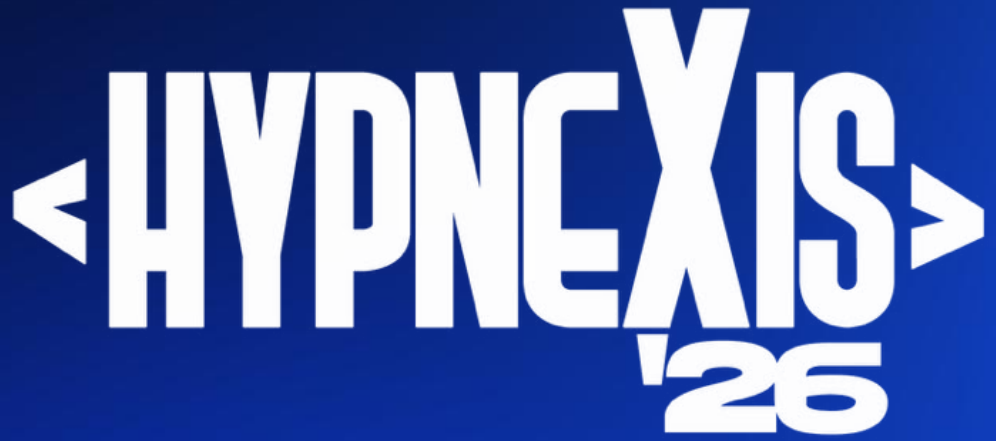
Implemented stack

- React: https://react.dev/
- TypeScript: https://www.typescriptlang.org/
- Vite: https://vite.dev/
- Express: https://expressjs.com/
- Three.js: https://threejs.org/
- AlaSQL: https://github.com/AlaSQL/alasql
- SheetJS / XLSX: https://docs.sheetjs.com/
- pdf-parse: https://www.npmjs.com/package/pdf-parse

Optional integrations

- Google GenAI SDK: https://ai.google.dev/gemini-api/docs
- OAuth provider documentation supports server-side redirect configuration; real credentials were not present in this environment.

No benchmark or production-scale performance claim is made in this deck.



SYNORA.

Evidence-backed analysis. More trustworthy decisions.

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